Projectives in $\mathcal{O}$

① Enough projectives:
- cover simple L_μ
- use lifting etc to cover any module via composition ex. $P' \oplus P$ (sketch)

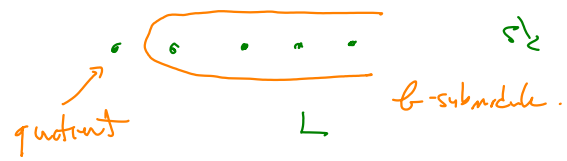
Find λ dominant s.t. $\lambda-\mu$ integral + dominant

$$\Delta_\lambda \otimes L_{\lambda-\mu} \twoheadrightarrow \Delta_\mu \twoheadrightarrow L_\mu \qquad 0 \to L_\lambda \to M \twoheadrightarrow L_\mu \to 0$$

projective b/c λ dom.

finite dim

standard filtration $\Delta_\lambda \otimes L$ (f.d.) use $\mathfrak{b}$-filtration on L i.e. "lowest weight is quotient"



General artinian module theory

- Every simple has a 'minimal' projective cover

i.e. $P_\lambda \twoheadrightarrow L_\lambda$ s.t. no submodule of P_λ surjects onto L_λ

this implies that $P \twoheadrightarrow L_\lambda$. then P_λ must be a summand.

surjective, $\operatorname{im}(\varphi) \subseteq P_\lambda$ which surjects onto $= P_\lambda$ $\Rightarrow$ P_λ is summand.

- P_λ indecomposible.
- In artinian category, projective covers exist, and projective covers of simples are the only indecomposible projectives.

P projective, $P \twoheadrightarrow L_\lambda$ pull off P_λ, repeat.

Remark: Dually, injective envelopes $L_\lambda \hookrightarrow I_\lambda$ (worst) ∇_λ costandard

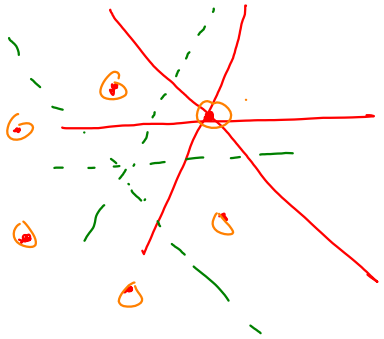
- Projective have standard filtrations
  - By our construction of P_λ as quotient of $M_\mu \otimes L_{\mu-\lambda}$
    "worst" the standard filtration.

· (Morita theory) $\lambda \in \mathfrak{h}^*$ $\quad P = \bigoplus_{w \in W} P_{w\cdot\lambda}$

P is projective generator, and

$$\mathrm{Hom}(P, -) : \mathcal{O}_{[\lambda]} \xrightarrow{\sim} \mathrm{End}(P)^{op}\text{-modules}$$

$\mathfrak{sl}_3$: $\mathcal{O}_0$



$P_0 = \Delta_0$

$P_{-\alpha_1}$

$P_{-\alpha_2}$

$P_{-2\alpha_1-\alpha_2}$

$\vdots$

$\mathfrak{sl}_2$: Classify all indecomposable modules in $\mathcal{O}_0$

Possible objects: $L_0 \quad L_{-2} \quad \Delta_0 \quad \nabla_0 \quad P_{-2}$

$L_{-2} = \Delta_{-2} = \nabla_{-2}$, $\quad \Delta_0 = P_0$, $\quad \nabla_0 = I_0$, $\quad P_{-2} = I_{-2}$

self-dual projective/inj.
has evg standard mult 1
costandard mult 1

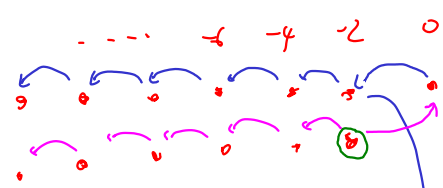
$$0 \to \Delta_0 \hookrightarrow P_{-2} \twoheadrightarrow \Delta_{-2} \to 0$$
$$\| $$
$$0 \to \nabla_{-2} \hookrightarrow I_{-2} \twoheadrightarrow \nabla_0 \to 0$$

$L_{-2} \quad L_0 \quad L_{-2}$

$$0 \subseteq L_{-2} \subseteq \Delta_0 \subseteq P_{-2} \subseteq 0$$

Remark
generated by non-highest weight

Δ_0



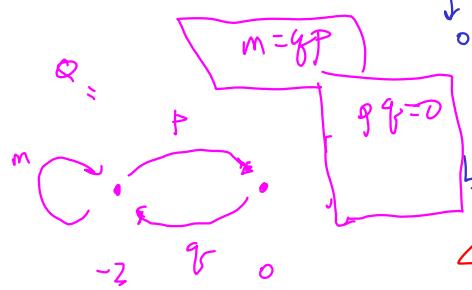
DModule
$\mathbb{A}^1$
D/xdx

How to show these are all decomposables?

Morita thm: $P = P_{-2} \oplus P_0$

· $\mathrm{End}(P)$ quiver algebra

Q:



$P_{-2} \twoheadrightarrow L_{-2} \hookrightarrow P_{-2}$

$\mathrm{Hom}(P_{-2}, P_{-2})$: 2

Δ_0: $P_0 \to P_{-2}$ 1

$P_{-2} \to P_0$ 1

Δ_0: $P_0 \to P_0$ 1

$P_{-2} \twoheadrightarrow L_{-2} \hookrightarrow \Delta_0$

$\mathrm{Hom}(P_1, -) : \mathcal{O}_0 \xrightarrow{\sim} Q^{op}\text{-mod}$

L_{-2}

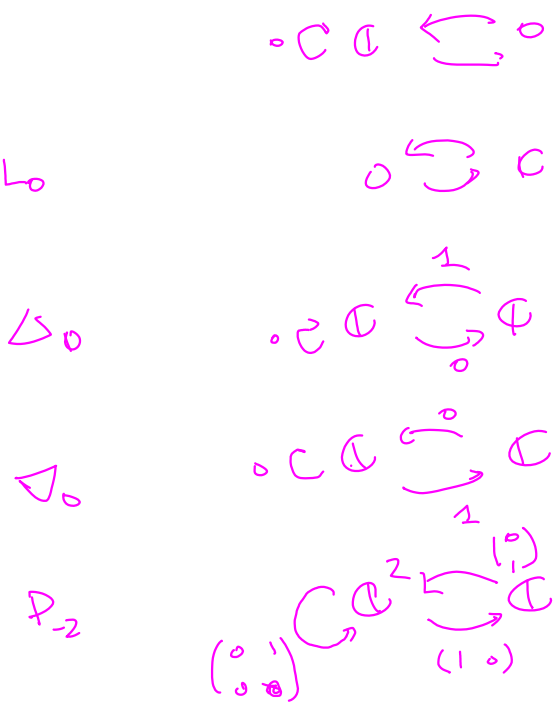
$$\begin{tikzcd} \mathbb{C} \arrow[r, bend left, "p^*"] & \mathrm{Hom}(P_2, L_{-2}) \arrow[l, bend left, "q^*"] & \mathrm{Hom}(P_0, L_{-2}) = 0 \end{tikzcd}$$

L_0

Δ_0

∇_0

P_{-2}



<u>Exercise</u>: Show that these are all indecomposables of the quiver Q, w/ relations

$\hookrightarrow$ i.e. an object of $\mathcal{O}_0$

$\mathrm{Ext}^{\infty}(T) = 0$.

"tilting objects"

P_{-2}

$\rightsquigarrow$ derived equiv.

$\cdots$

s.t. $QP = 0$.

Geom. geometric interpretation of this quiver.

$D^b\mathrm{Mod}(\mathbb{P}^1) \xrightarrow{RH} \mathrm{Sh}(\mathbb{P}^1)$

$\mathcal{F}_{(0,\varepsilon)} \longrightarrow \mathcal{F}_1$

$[1]$

tw. t-mon.

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